Elon Musk Twitter Analysis

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1 Introduction

Since the dawn of the digital age, social media has played a pivotal role in the decision making of people across the globe. Companies have invested millions of dollars in social media marketing, enabling them to utilize real-time insights to make data-driven decisions. Because of this overlap between social media and business marketing, the relationship between social media and the financial market has also become an increasingly interesting topic. With the growing influence of individuals such as Elon Musk who can directly influence market sentiment through unregulated social media posts, questions have been raised regarding the effect of these posts on the highly regulated capital market. Traditional finance theories, such as the Efficient Market Hypothesis (EMH), argue that asset prices already reflect all available information, making them unpredictable and immune to new public disclosures [7]. However, this view has come under fire in recent years as empirical evidence suggests that social media posts, especially from influential individuals, directly influence market sentiment, therefore contradicting the assumptions of perfect market efficiency. Although social media began as a communication tool to connect friends and family across the globe, it has quickly transformed into a powerful influence mechanism across domains like policymaking, business, and finance [1] [2]. Looking at financial contexts specifically, Kumar and Devi (2014) [4] argue that social media supports real-time sentiment diffusion and shapes investor behavior. This is especially applicable in the case of Elon Musk, whose tweets have been shown to directly impact financial and crypto markets. Roška and Špoljarić (2024) [10] found that Musk's tweets significantly impact both Bitcoin and Tesla stock, with the magnitude of these effects growing alongside his public prominence. As Elon transitioned from a revolutionary tech entrepreneur, to the richest man in the world, to eventually being a part of the current presidential administration, his effect on the market directly reflected his growing influence. Looking specifically at crypto markets, Shahzad et al. (2022) [11] reinforce this finding, showing that his posts often result in price explosiveness in cryptocurrency markets. Furthermore, Metta, Madhavan, and Narayanan (2022) [6] investigate the "Power of 280" and demonstrate that Musk's tweet characteristics (length, timing, and sentiment) can significantly move stock prices, particularly

Tesla's. Looking at Elon Musk's tweets on a case by case basis, Kuokka (2021) [5] provides evidence that Musk's social media activity correlates with irregular stock returns in U.S. markets, identifying him as a unique driver of market sentiment. Even when conducting a broader review of social media influence on the stock market, Neri-Mares et al. (2024) [8] highlight Musk as a key example of how sentiment-rich posts on platforms like Twitter can drive short-term volatility, particularly in growth-oriented firms. These findings collectively pave the way for a data-driven approach to truly understand Elon's effect on the market.To address this gap, we implement three distinct machine learning models to explore different aspects of Musk's social media influence on Tesla Stock. Sentiment Impact Model: Drawing on prior research that associates tweet tone with market reaction [6] [11] our project assesses how positive vs negative sentiment in Musk's tweets corresponds to changes in Tesla stock price. Time-Series Analysis Model: Inspired by Roška and Špoljarić (2024)[10] and Kuokka (2021)[5], this model evaluates how the impact of Musk's tweets has evolved over time, comparing Tesla's early years, growth phase, and recent maturity to see whether market sensitivity has increased with his influence. Keyword Analysis Model: Drawing inspiration from Neri-Mares et al. (2024)[8] and Poyser (2017)[9], we apply machine learning to detect whether specific words or themes in Musk's tweets consistently correlate with market behavior.

Together, these models provide us with a wide breadth of analysis that will help us quantify the relationship that prior studies have explored. This project aims to gain a deeper understanding of how social media can translate into tangible financial outcomes.

2 Importance/Significance of Project

The rise of social media in the past few decades has opened the door for those with immeasurable power to take advantage of their influence and sway human decision making which therefore impacts financial markets. Elon Musk is a perfect example of this as his tweets have shown to directly influence both Tesla's stock price and the cryptocurrency market. As shown in studies like Roška & Špoljarić (2024) [10], Metta et al. (2022),[6] and Shahzad et al. (2022)[11], Musk has a direct impact on the market which contradicts the traditional financial theories such as the Efficiency Market Hypothesis outlined in Mishkin et al., 2005[7]. From a financial perspective, this research offers insights that may help investors interpret or respond to tweet-driven volatility. In fact, this problem may even have regulatory implications regarding market manipulation. From an academic standpoint, this problem provides a foundation to develop new economic models based on non-traditional market drivers, replacing the Efficiency Market Hypothesis.

3 Challenges of this Problem

Drawing a line between correlation and causation in this problem is extremely important. Just because a tweet and a stock price change occur close in time does not imply causation. Given that Musk's tweets are often sarcastic in nature, it may make sentiment analysis extremely difficult. To find a correlation between tweets and stock data, extensive preprocessing of data is essential to reaching comprehensive conclusions.

4 Clear Scope of the Problem

This project is clearly scoped into three different machine learning models to find a relationship between Elon's tweets and Tesla stock changes.

- Sentiment Impact Model: Assess how positive vs negative sentiment in Musk's tweets corresponds to changes in Tesla stock price.
- Time-Series Analysis Model: Evaluates how the impact of Musk's tweets has evolved over time, comparing Tesla's early years, growth phase, and recent maturity to see whether market sensitivity has increased with his influence.
- Keyword Analysis Model: Apply machine learning to detect whether specific words or themes in Musk's tweets consistently correlate with market behavior.

A solution to this problem includes:

- A predictive model that shows statistically significant associations (or lack thereof) between tweet features and stock price changes.
- A timeline-based analysis that reveals how Musk's market influence has evolved.
- An interpretable system that helps isolate impactful tweet elements.

5 Limitations of this problem

Given the volatility of the stock market, being able to directly predict stock performance from Musk's tweets is nearly impossible. Instead, we hope to derive a causality relationship based on historical data but it is impossible to isolate the stock from all extraneous variables. To address this problem most effectively, an understanding of time-series analysis, keyword extraction, and sentiment analysis are crucial.

5.1 Data

We began looking for data related to Elon Musk's tweets and Tesla stock prices. We settled on two kaggle datasets, one for each of our requirements. The Elon Musk tweets dataset had all of his tweets from 2010 to 2025. The important fields taken from this were the data and original tweet text. This dataset also included other data, including count, view count, retweet count etc. But these fields were often NaN and this is likely because some of the tweets were either deleted or are too old, but nonetheless many of these values were unhelpful. The tesla stock data that we found had fields like date, open, high, low, close etc. We found that this dataset satisfied all of our requirements.

We split up the dataset into three distinct periods of time. An early period from 2012 to 2017. A middle period from 2017 to 2019. And a dense period which was the rest. We did this because Elon did not tweet as much in twitter's early days and his popularity also was not as high. With this method, we are able to make predictions for each individual time period and see which one has the best performance. Now a problem that we had with the dataset is that it didn't indicate if a tweet had a negative or positive impact on the Tesla prices. Because of this we had to figure out a method to either inference this or compute this. One such method was to find keywords from Elon's tweets and correlate it with the stock price. We used a time lag

of 1,2 and 3 on the stock price to see how the tweet may have affected the price of the stock for the following days. By doing this we grouped positive and negative keywords for each period.

6 Methodologies

6.1 Keyword method

This method could be considered the simplest method, which is good to use a baseline to evaluate other methods. In our keyword method, we combined the datasets together based on time. We then cleaned the text, removing any superficial data such as urls, special characters, etc. We then used VADER for high frequency keywords to label them as positive or negative. We then used a supervised binary classification problem to predict the Tesla stock price increases and decreases for 1,2, and 3 days after an Elon Musk tweet. We trained 9 models by doing a combination of 3 time periods, early middle and dense with 3 lags 1,2 and 3 days. The training methodology and fine tuning was that we split our dataset into an 80-20 split of train and test data with a batch size of 16 and 25 epochs with the adam optimizer and crossentropyloss. We adjusted our learning rate and tried to ensure that our models did not have overfitting or underfitting.

6.1.1 Conclusion

For the keyword method, we found all combinations had an accuracy between 40 to 60 percent for all methods; recall was almost perfect for many increases but incredibly bad for decreases in stock price. This can be explained by the fact that for the most part from 2010 to 2025, Tesla stock prices increased with small decreases every once in a while, but the general trend is that it increased. We believe that the volatility of both the stock market and Elon musk's tweets lead to a very difficult problem for a model to predict. The stock market is impacted by a multitude of factors that cannot begin to be described from a single dataset. There are just too many variables to consider and to understand which variables actually matter on which certain day is near impossible. Elon Musk is also very random in how he tweets. The general way he speaks changes depending on his topic and he is often very sarcastic. This sarcasm does not translate well for a machine learning model.

6.2 Sentiment Analysis Method

For the Sentiment Analysis portion, we implemented a lexicon-seeded weak-supervision approach in which we used VADER to assign "seed" labels to tweets (compound ≥ 0.2 as positive, ≤ -0.2 as negative), embedded the full corpus via TF-IDF with 5000 features, and ran KMeans ($k = 2$) to propagate binary labels across all posts. We then calculated each day's fraction of positive tweets and plotted that positive-rate against next-day TSLA returns in a scatter plot (Figure 1).

Our results from the lexicon-seeded pipeline showed a Pearson correlation of $r = 0.004$ between daily positive-tweet rate and next-day return. The scatter plot forms a dense cloud centered at zero, revealing no discernible linear trend or clustering that would suggest sentiment drives price changes.

We then examined continuous VADER metrics without binarization by aggregating the daily mean and standard deviation of compound scores. We produced scatter diagrams of mean score versus return and volatility versus return to visualize any linear relationships (Figure 2).



Figure 1: Daily mean sentiment vs. next-day return (full-page view).

These continuous VADER analyses yielded correlations of r_mean = 0.016 and r_vol = 0.017, and the scatter diagrams remain a diffuse swarm around the origin. There is no apparent slope or directional bias suggesting that either average sentiment or its daily volatility relates to next-day returns.

Next, we compared the distribution of next-day open–close price differences by assigning days as positive or negative based on the weak-labels and generating boxplots, swarm plots, and density histograms (Figures 3, 4, 5) to assess differences in central tendency and tail behavior.

The distributional comparison showed that positive-labeled days (n = 9071) have a mean open–close difference of 0.853$, SD = 8.27$, whereas negative-labeled days (n = 1243) have a mean of 0.624$, SD = 7.64$. A Welch's t-test yields t = 0.981 with p = 0.327, confirming no statistically significant difference and heavily overlapping distributions in all visualizations.

We then moved on to a transformer-based pipeline using DistilBERT (SST-2) to score each cleaned tweet with a positive-class probability. We averaged these probabilities per day and plotted the resulting daily mean BERT score against next-day returns in a correlation scatter plot (similar in style to Figure 1).

The transformer-based correlation analysis

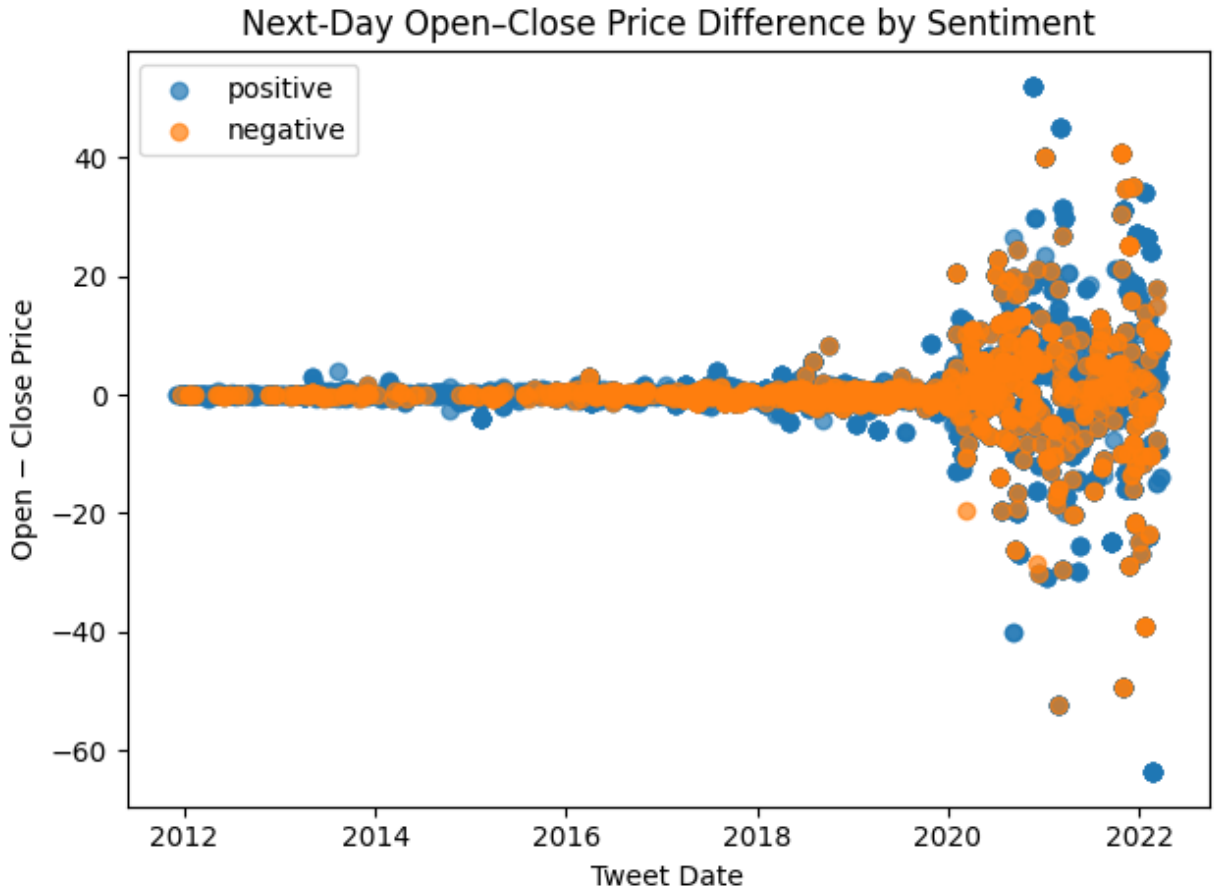


Figure 2: Tweet date vs. Open-Close Price (full-page view).

returned a Pearson r = 0.010 between daily mean BERT sentiment and next-day returns, with the scatter plot again forming a cloud centered on zero, indicating negligible association.

To evaluate predictive power, we fit a linear regression model using daily mean BERT sentiment as the sole predictor for next-day returns and assessed model performance via R^2 and mean squared error.

The linear regression yielded $R^2 \approx -0.002$ and MSE ≈ 0.0021, indicating that the sentiment predictor explains no variance beyond the baseline mean model and performs worse than simply predicting the average return.

We then trained a Random Forest classifier with the daily mean BERT score as the single feature to predict whether next-day returns would be positive or negative, using five-fold time-series split cross-validation for accuracy assessment.

The Random Forest achieved a mean CV accuracy of approximately 53.2% and an in-sample accuracy of about 58.6%, demonstrating only near-chance predictive performance, despite assigning full feature importance to the sentiment variable.

Finally, we explored time-series dynamics by computing weekly median open–close differences for days labeled positive versus negative and by smoothing these series with a three-month rolling median. We plotted both raw weekly medians and the smoothed curves to observe any longer-term diver-

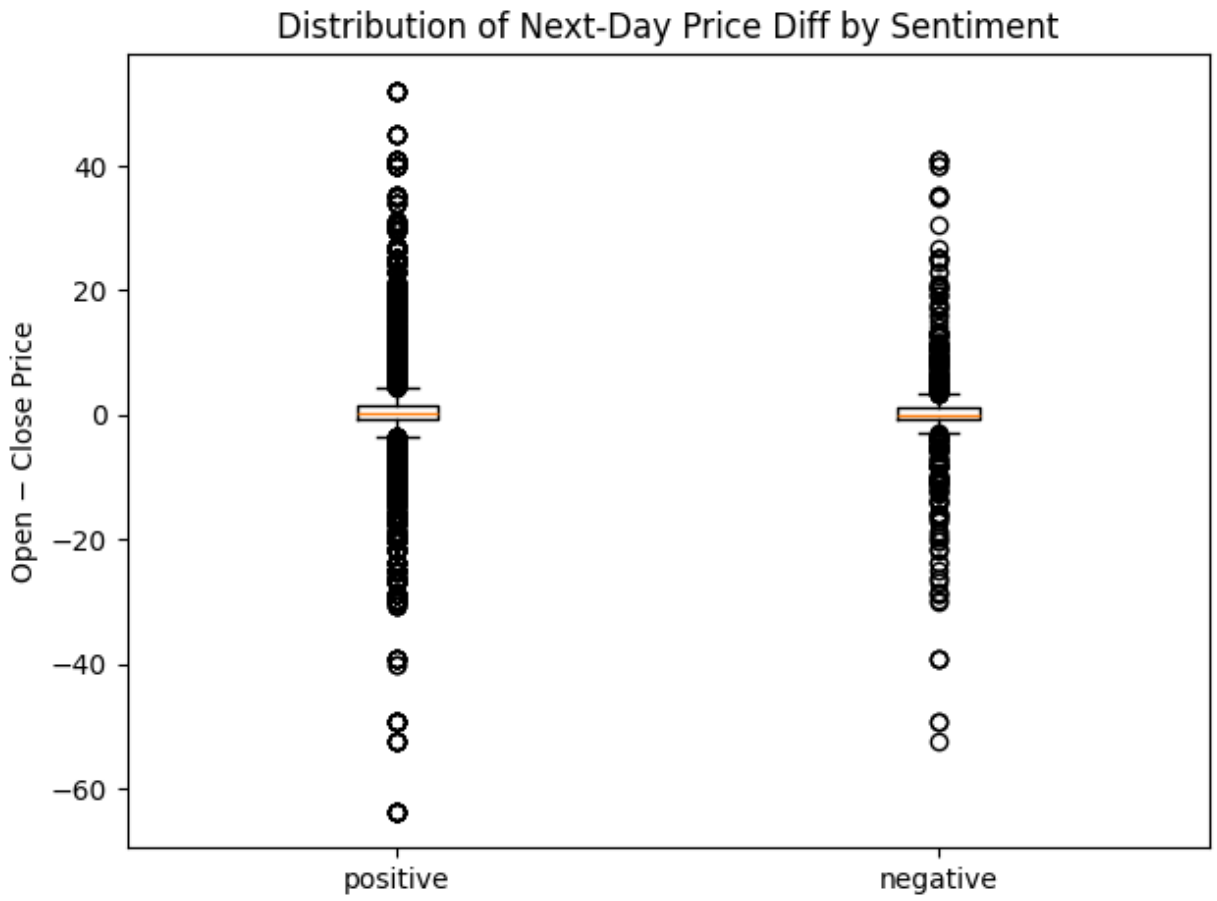


Figure 3: Distribution of Next Day Price Difference (full-page view).

gences (Figures 6, 7).

Both the weekly median series and the three-month rolling medians for positive and negative sentiment classes move almost in lockstep from 2010 through 2022, with no sustained divergence even during periods of elevated market volatility

6.2.1 Conclusion

After conducting lexicon-seeded weak supervision, continuous VADER scoring, distributional comparisons, transformer-based correlation, linear regression, Random Forest classification, and week- and quarter-level rolling median analyses, and examining scatter plots, boxplots, density histograms, and smoothed time series, we found no robust evidence that Elon Musk's tweet sentiment meaningfully predicts next-day Tesla stock movements. We therefore conclude that tweet polarity in our sample does not carry exploitable information for forecasting TSLA returns.

6.3 Analysis

The keyword-based classification approach, while conceptually simple, provided a valuable baseline for evaluating more sophisticated methods. By aligning tweets with Tesla stock performance over different time lags and time periods, and labeling tweets based on keyword sentiment using VADER,

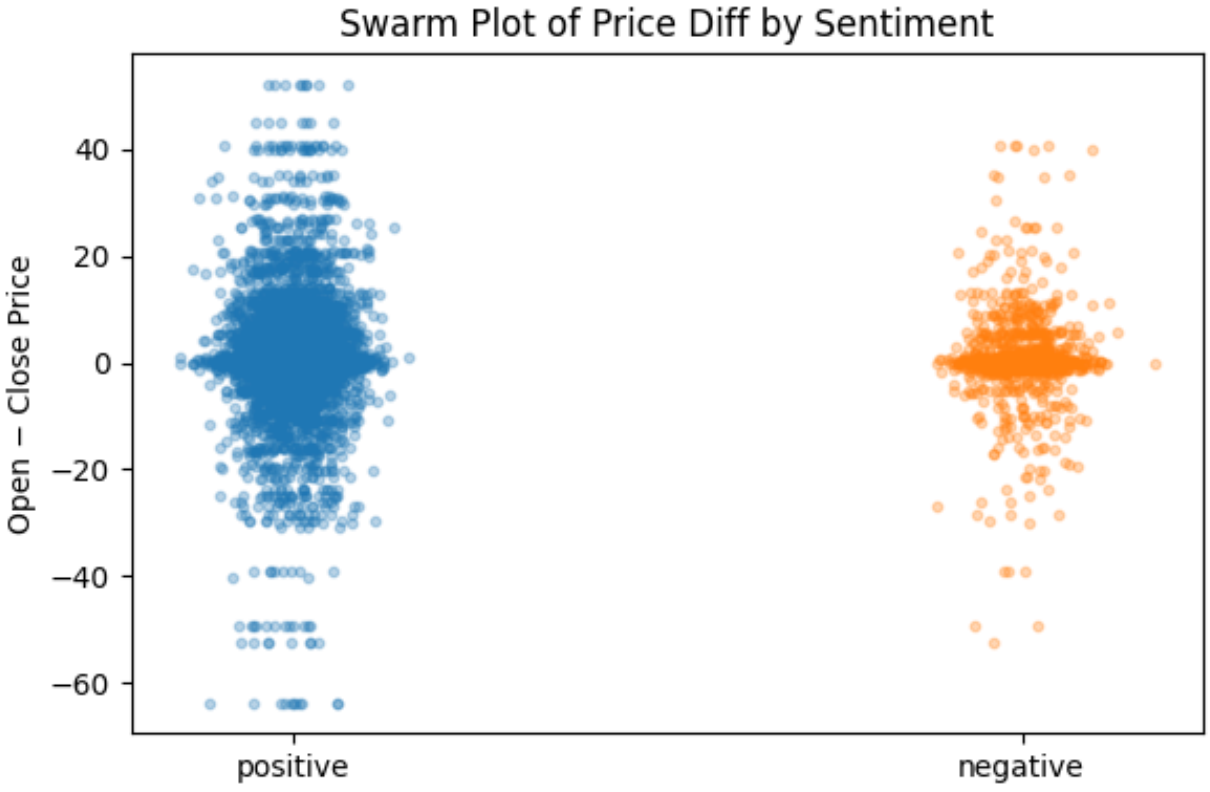


Figure 4: Swarm plot of Price Difference (full-page view).

we established a structured binary classification framework. Although our training process was thorough, employing standard splits, optimization strategies, and multiple epochs, the models demonstrated mediocre performance, with accuracy generally hovering between 40–60%. This inconsistency was most evident in the recall imbalance, where the models were much more successful in identifying stock increases than decreases. This bias is likely a reflection of the broader historical trend in Tesla stock, which saw steady upward growth over the examined period, making decreases both rarer and harder to predict.

These findings suggest that the keyword method, though useful for structuring the prediction task, lacks the nuance to capture the complex relationship between tweet content and stock price fluctuations. The model's inability to predict downward trends highlights an important limitation: tweets, especially from a figure as erratic as Elon Musk, do not convey sentiment in a way that easily maps onto binary financial outcomes. Sarcasm, ambiguity, and topic shifts often muddle the sentiment signal, and without deeper contextual understanding or temporal nuance, models based on keyword frequencies struggle to perform meaningfully. This reinforces the necessity of more context-aware models that move beyond surface-level sentiment, especially when attempting to extract predictive insight from public discourse.

6.4 Final Thoughts

Through our analysis, we found that there is no strong or consistent correlation between Elon Musk's tweets and Tesla's stock

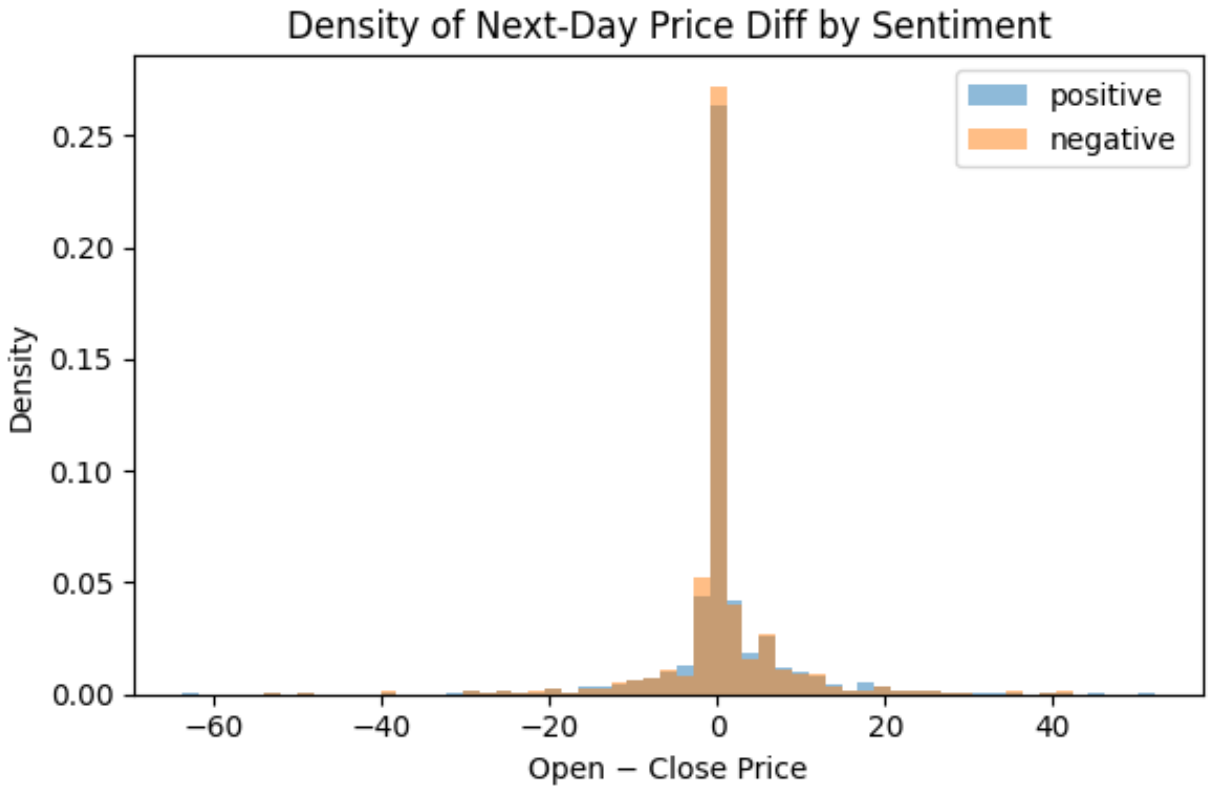


Figure 5: Density of Next day Price Difference (full-page view).

price movements. Despite his strong public influence, our results suggest that isolating the effect of Musk's tweets on Tesla stock is extremely challenging, likely due to the presence of numerous extraneous factors that simultaneously influence stock prices. These factors include broader market conditions, company financials, macroeconomic news, regulatory developments, and investor sentiment, which collectively overshadow the impact of any single person, even a high-profile figure like Musk. However, an interesting observation from our study is the presence of a weak association between the timing of Musk's tweets and general Tesla stock movement, regardless of whether the movement was positive or negative. This finding, while not strong enough to claim causality, opens the door for further investigation into whether Musk's tweets serve more as signals for market activity. It is also possible that our machine learning models may not have fully captured the nuances of Musk's influence, particularly given the complexity of sentiment, sarcasm, timing, and market context embedded in his tweets. Given these findings, there is potential value in replicating this study using alternative modeling approaches, including deep learning methods that are better equipped to parse linguistic nuance, sarcasm, and subtle emotional cues. Looking forward, expanding the dataset, including a longer time horizon, more granular trading data, or combining Tesla's stock movement with intraday tweet timing, may yield more meaningful results. Another promising direction is to analyze the impact of Musk's tweets on the cryptocurrency market, which is less regulated and often more reactive to social me-

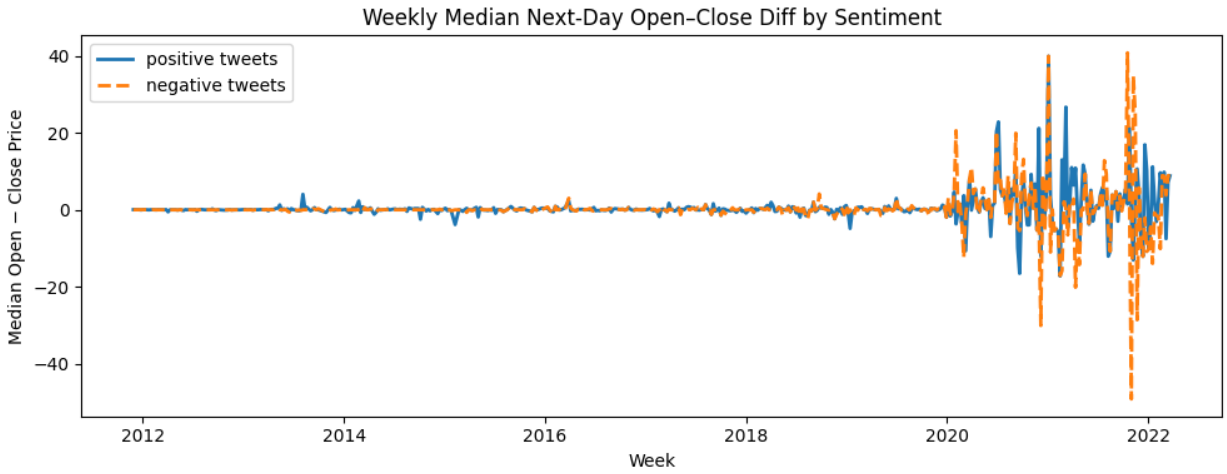


Figure 6: Weekly Median Next Day Open Close Difference (full-page view).

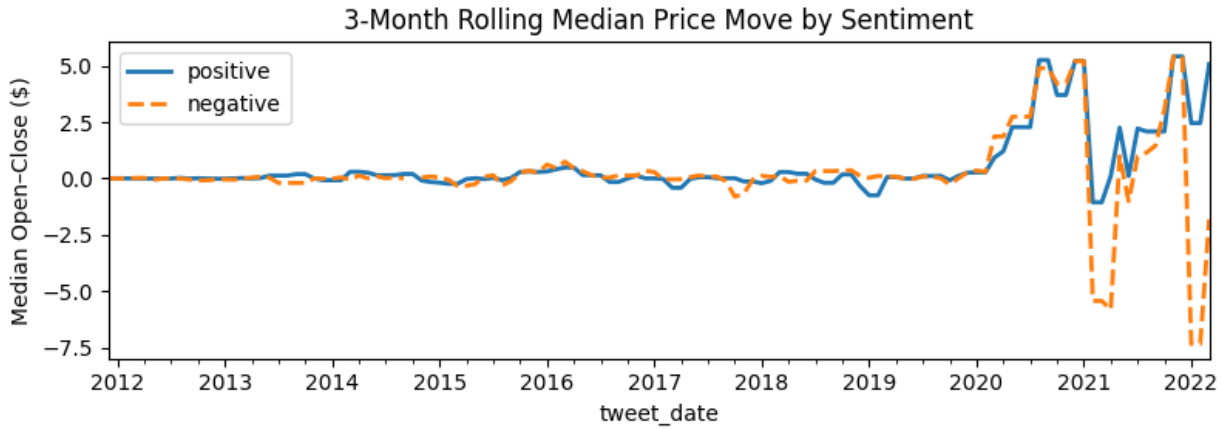


Figure 7: Three Month Rolling Median Price Move (full-page view).

dia signals. Since Musk has previously influenced major crypto assets like Bitcoin and Dogecoin, studying these reactions may offer clearer insights into the power of social media in less institutionally anchored markets. Finally, future research could explore whether Musk's tweets correlate with broader market indices or investor sentiment at large, rather than Tesla stock alone. In summary, while our current models did not reveal a strong link between Musk's tweets and Tesla's stock price, our findings reinforce the complexity of attributing financial outcomes to singular social media events. This project lays the foundation for continued interdisciplinary exploration at the intersection of finance, communication, and machine learning.

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